Package 'cfisher'

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Type Package

Title Fisher's Exact Test Controlling for Confounders

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Description Tests the association between a binary exposure and a binary outcome while controlling for confounders, in the manner of Fisher's exact test. A logistic model of the outcome on the confounders, without the exposure, gives each subject's probability of being a case under the null hypothesis. The test then uses the exact null distribution of the number of exposed cases, either given the exposed subjects and the number of cases (a generalization of the hypergeometric distribution of Fisher's exact test) or given the exposed subjects only (a Poisson-binomial distribution). It returns the p-value, an estimate of the odds ratio and its confidence interval. With confounders the fitted model is plugged in, so the p-value and the interval are exact at the true confounder effects. Without confounders the results agree with 'fisher.test()'. The null model can be fitted once and reused, which suits scans of many exposures, such as genetic variants, against one outcome. The method is described in a draft manuscript by Wu and Contador (2026), Fisher's exact test controlling for confounders.

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`cfisher-package` *cfisher: Fisher's Exact Test Controlling for Confounders*

Description

Tests the association of a binary exposure with a binary outcome while controlling for confounders, in the manner of Fisher's exact test. The main function is `cfisher.test`. `cfisher.null` returns the null distribution of the number of exposed cases that the test uses. The method is described in the paper of Wu and Contador (2026), which is a draft.

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References

Wu, Z. and Contador, G. (2026). Fisher's exact test controlling for confounders. Draft manuscript.

`cfisher.null` *Null distribution of the number of exposed cases*

Description

Returns, as a data frame, the null distribution of T, the number of cases among the exposed subjects, that `cfisher.test` uses. Other packages can take it as a discrete null distribution (a support and its probabilities).

Usage

```
cfisher.null(x, y, z = NULL, conditional = TRUE, fit = NULL)
```

Arguments

`x`	Integer or logical vector of 0/1 exposure values, one per subject.
`y`	Integer or logical vector of 0/1 outcomes (1 = case), the same length as `x`.
`z`	Confounders: `NULL` (the default; the null model has an intercept only, so the test is Fisher's exact test), or a numeric vector, matrix or data frame with one row per subject. An intercept is always added. Ignored when `fit` is given.
`conditional`	Logical. `TRUE` (the default): the null distribution is the conditional Bernoulli distribution of T given the exposed subjects and the number of cases, which is Fisher's exact test controlling for confounders. `FALSE`: the null distribution is the Poisson-binomial distribution of T given the exposed subjects only, the unconditional test (the null of the BinomiRare test and of the conditional randomization test).

`fit` Optional null probabilities, which take precedence over `z`: a fitted `glm` object of the binomial family whose fitted values are the probabilities that each subject is a case under the null hypothesis, or a numeric vector of such probabilities, strictly between 0 and 1, one per subject. Passing `fit` lets a scan over many exposures fit the null model once. The model must not contain the exposure. For the odds ratio, a `glm` must have the logit link and no weights or offset, and its model matrix is used to refit the slopes; a vector of probabilities allows no refit (see Details).

Details

Let m be the number of exposed subjects, r the number of cases and μ_i the fitted null probability that subject i is a case, with odds $w_i = \mu_i/(1-\mu_i)$.

With `conditional = TRUE` the distribution is the conditional Bernoulli distribution of T given the exposed set and r:

$$P(T = t) = e_t(w_E)e_{r-t}(w_{E^c})/e_r(w),$$

where E is the exposed set and e_k is the elementary symmetric function of order k of the odds in a set. It is computed as the ratio of Poisson-binomial probabilities

$$P(T = t) = f_E(t)f_{E^c}(r-t)/\sum_s f_E(s)f_{E^c}(r-s),$$

where f_A is the probability mass function of the number of cases in the set A, by the forward recursion over subjects, in $O(m^2 + (n-m)r)$ time, so that no elementary symmetric function is formed. The distribution does not change when every odds is multiplied by a constant, so before the recursion the log-odds are shifted by the constant that makes $\sum_i \mu_i = r$. This keeps the recursion at the center of the distributions; for a fitted model with an intercept the shift is zero. If all odds are equal the distribution is hypergeometric, and if the odds take one value on the exposed and another on the unexposed it is Fisher's noncentral hypergeometric distribution.

The recursion runs in probability scale, which is fast, as long as a bound on its error from underflow is below 10^{-12} of every probability it has to deliver (roughly, of those above 10^{-290}). Otherwise it runs in log scale, at about 20 times the cost, so that no probability is lost to underflow, even far below the range of double precision. In both scales the probabilities carry the rounding error of double-precision arithmetic, a relative error that grows with the number of subjects.

With `conditional = FALSE` it is the Poisson-binomial distribution of T given the exposed set, the distribution of a sum of m independent Bernoulli variables with probabilities μ_i, $i \in E$, computed in $O(m^2)$ time. The number of cases is left random.

Assumptions: given the confounders, the outcomes are independent Bernoulli variables with the null probabilities μ_i (the outcome does not depend on the exposure given the confounders). The fitted probabilities are plugged in as if known.

Value

A data frame with columns `support` (the possible values of T, in increasing order) and `prob` (their probabilities, summing to one). A probability below the range of double precision (about 10^{-308}) is shown as 0 or with reduced precision; `cfisher.test` works with the logarithms, which keep such probabilities.

References

Wu, Z. and Contador, G. (2026). Fisher's exact test controlling for confounders. Draft manuscript.

Chen, S. X. and Liu, J. S. (1997). Statistical applications of the Poisson-binomial and conditional Bernoulli distributions. *Statistica Sinica*, 7(4), 875-892.

See Also

cfisher.test

Examples

```
set.seed(1)
n <- 200
z <- rnorm(n)
x <- rbinom(n, 1, plogis(-2 + z))
y <- rbinom(n, 1, plogis(-1 + z))
cfisher.null(x, y, z)
cfisher.null(x, y, z, conditional = FALSE)
```

---

cfisher.test *Fisher's exact test controlling for confounders*

Description

Tests whether a binary exposure x is associated with a binary outcome y while controlling for confounders z, in the manner of Fisher's exact test. The statistic is T, the number of cases among the exposed subjects. Its null distribution is exact, given a logistic model of the outcome on the confounders that does not contain the exposure. Without confounders the test reproduces fisher.test on the 2×2 table of x and y.

Usage

```
cfisher.test(
  x,
  y,
  z = NULL,
  alternative = c("two.sided", "greater", "less"),
  conf.int = TRUE,
  conf.level = 0.95,
  midp = FALSE,
  conditional = TRUE,
  fit = NULL
)
```

Arguments

x	Integer or logical vector of 0/1 exposure values, one per subject.
y	Integer or logical vector of 0/1 outcomes (1 = case), the same length as x.
z	Confounders: NULL (the default; the null model has an intercept only, so the test is Fisher's exact test), or a numeric vector, matrix or data frame with one row per subject. An intercept is always added. Ignored when fit is given.
alternative	Character. "two.sided" (the default), "greater" (the exposed are enriched among the cases, p-value $P(T \geq t)$) or "less" ($P(T \leq t)$).

`conf.int` Logical. Whether to compute the estimate and the confidence interval of the odds ratio. If `FALSE` both are omitted, which is faster when many tests are run. (Unlike `fisher.test`, which still returns its estimate, the omission of the estimate here is intentional, for speed in scans.)

`conf.level` Confidence level of the interval, strictly between 0 and 1, such that each tail probability of the interval lies between 10^{-12} and $1 - 10^{-12}$.

`midp` Logical. If `TRUE` the observed count carries half its probability in the p-value (the mid-p value). Only the p-value changes; the estimate and the confidence interval are those of the exact test.

`conditional` Logical. `TRUE` (the default): the null distribution is the conditional Bernoulli distribution of T given the exposed subjects and the number of cases, which is Fisher's exact test controlling for confounders. `FALSE`: the null distribution is the Poisson-binomial distribution of T given the exposed subjects only, the unconditional test (the null of the BinomiRare test and of the conditional randomization test).

`fit` Optional null probabilities, which take precedence over `z`: a fitted `glm` object of the binomial family whose fitted values are the probabilities that each subject is a case under the null hypothesis, or a numeric vector of such probabilities, strictly between 0 and 1, one per subject. Passing `fit` lets a scan over many exposures fit the null model once. The model must not contain the exposure. For the odds ratio, a `glm` must have the logit link and no weights or offset, and its model matrix is used to refit the slopes; a vector of probabilities allows no refit (see Details).

Details

Let E be the exposed subjects, $m = |E|$, r the number of cases and μ_i the null probability that subject i is a case, the fitted value of the logistic regression of `y` on `z` (with intercept, without `x`), with odds $w_i = \mu_i/(1 - \mu_i)$.

Null distribution. Given the confounders and the exposure, the null hypothesis makes the outcomes independent Bernoulli(μ_i) variables. The exposed set is held fixed. With `conditional = TRUE` the number of cases r is held fixed too, and T has the conditional Bernoulli distribution

$$q(t) = e_t(w_E)e_{r-t}(w_{E^c})/e_r(w),$$

with e_k the elementary symmetric function of order k of the odds in a set. It is the hypergeometric distribution when all odds are equal, and Fisher's noncentral hypergeometric distribution when the odds take one value on the exposed and another on the unexposed. It is computed as a ratio of Poisson-binomial probabilities by a forward recursion over subjects in $O(m^2 + (n - m)r)$ time (see `cfisher.null`). With `conditional = FALSE`, T has the Poisson-binomial distribution of the exposed subjects' probabilities.

P-values. The one-sided p-values are $P(T \geq t)$ and $P(T \leq t)$. The two-sided p-value is the sum of the probabilities of the support points that are no more probable than the observed one, the convention of `fisher.test`; two probabilities are treated as tied when they agree to a relative tolerance of 10^{-7}, the tolerance of `fisher.test`. The mid-p value subtracts half the probability of the observed count.

Odds ratio. The exposure enters the logistic model with coefficient β, so that e^β is the odds ratio of the exposure given the confounders. For each hypothesized β the slopes of the confounders are refitted by maximum likelihood with βx_i as an offset, the null distribution of T is computed at the resulting odds (conditional or Poisson-binomial, as chosen), and the test is inverted. The estimate solves $E_\beta(T) = t$. The lower limit solves $P_\beta(T \geq t) = \alpha/2$ and the upper limit $P_\beta(T \leq t) = \alpha/2$,

where α is one minus `conf.level`; a one-sided interval puts all of α in one tail. The equations are solved for β in $[-40, 40]$ by root finding with an expanding bracket. A limit beyond that range is reported as `0` or `Inf`. The estimate is `0` or `Inf` when $E_\beta(T) - t$ is within 10^{-9} of zero at an end of that range, which is taken to mean that $E_\beta(T)$ reaches t only in the limit, as when the logistic model that contains the exposure has no finite maximum likelihood estimate. With `conditional = FALSE` and the slopes refitted, any other estimate beyond the range is found as the maximum likelihood estimate of β in the logistic model that contains the exposure, because $E_\beta(T) = t$ is then the profile likelihood equation of that model for β. Otherwise such an estimate is `NA` with a warning. Each refit starts from the nearest refit already computed, and a refit that does not converge makes the quantity being solved `NA`, with a warning. The refit runs on the columns of the design that the null fit keeps. The refitted tail probabilities need not be monotone in β, so the set of odds ratios that the test does not reject need not be an interval; the function warns when the values it evaluated show more than one crossing. Without confounders, the conditional test (the default) gives the estimate and the interval of `fisher.test`. Refitting is needed because the null model, fitted without the exposure, absorbs part of the exposure effect when the exposure is common, so tilting its distribution estimates the wrong family under an alternative. The refit is a plug-in: the p-value and the interval treat the fitted slopes as known, and they are exact at the true slopes. When the exposure is a linear combination of the intercept and the confounders, its effect is not identified: the function warns, and the estimate and the interval are `NA`. The coverage of the refitted interval was checked in a simulation reported in the accompanying paper (Wu and Contador, 2026, a draft; see References). When `fit` is a numeric vector, no refit is possible, and the distribution at β is the exponential tilt $q_\beta(t) \propto q(t)e^{\beta t}$ of the distribution at those probabilities. This is exact when they are the baseline probabilities under the alternative (the probabilities at $\beta = 0$ of the model that contains the exposure), and otherwise it is not recommended.

Numerical range. No probability of the null distribution is lost to underflow, at any point of the support $\max(0, r - n + m) \le t \le \min(m, r)$, including those far below the range of double precision (about 10^{-308}). The recursion runs in probability scale when a bound on its error from underflow is below 10^{-12} of every probability needed, and in log scale otherwise (see `cfisher.null`). Apart from underflow, the probabilities carry the rounding error of double-precision arithmetic, a relative error that grows with the number of subjects. The p-values and the tilt are computed from the log probabilities, so a tiny p-value keeps that precision down to about 10^{-308}, below which it is returned as 0 or with reduced precision. The distributions that the refit computes at each hypothesized odds ratio stay in probability scale, which is fast; the inversion uses only their tail probabilities at the level $\alpha/2$ and their means, for which the far tail that probability scale can lose is negligible.

Assumptions. Given the confounders, the outcomes are independent, and under the null hypothesis the outcome does not depend on the exposure. The fitted null probabilities are plugged in as if known, so with confounders the test is exact at the true probabilities and approximate at the fitted ones. The conditional distribution does not change when every odds is multiplied by a constant, so only the slopes of the null model matter to it, not its intercept. The exposure must take both values and the outcome too; otherwise the function stops with a message. A supplied probability of exactly 0 or 1 is an error. The function warns when some fitted probability is within 10^{-8} of 0 or 1, which suggests separation of the outcome by the confounders, as when a stratum has no cases or no controls. The subjects concerned are then nearly certain cases or controls and carry almost no information on the exposure. Under complete separation every subject is such, the null distribution of T is concentrated at the observed count, and the p-value carries no information. For the odds ratio, under separation the refitted model has no maximum likelihood estimate at any odds ratio, so the estimate and the interval are `NA` with a warning. A stratum without cases or without controls carries no information on the exposure, and the test on the data without such strata gives the odds ratio.

Value

An object of class "htest" with components

statistic	the observed count $T = t$ of exposed cases.
parameter	exposed (the number m of exposed subjects), cases (the number r of cases) and n (the number of subjects).
p.value	the p-value.
conf.int	a confidence interval for the odds ratio, with attribute conf.level. For the two-sided alternative each tail carries half of one minus conf.level, as in fisher.test; for the one-sided alternatives the interval is one-sided. A limit is 0 or Inf when the observed count is at an end of the support or when the limit lies beyond the search range (see Details), and NA (with a warning) when its root search fails or the effect is not identified. Omitted if conf.int = FALSE.
estimate	the estimate of the odds ratio, $\exp(\hat{\beta})$, where $\hat{\beta}$ solves $E_\beta(T) = t$; 0 or Inf when the observed count is at an end of the support or when $E_\beta(T)$ reaches t only in the limit (see Details), and NA (with a warning) when its root search fails, when it lies beyond the search range under the conditional distribution, when the observed count is certain at every odds ratio or when the effect is not identified. With conditional = FALSE it is named "odds ratio", because it is then the maximum likelihood estimate under the unconditional distribution. Omitted if conf.int = FALSE.
null.value	the odds ratio under the null hypothesis, 1.
alternative	the alternative.
method	a string naming the test, whether it is conditional and mid-p, and how the odds ratio was obtained (when it is computed), which is either "slopes refitted under each hypothesized odds ratio" or "tilt at the supplied odds".
data.name	the names of the data.
null.distribution	a data frame with columns support and prob, the null distribution of T that the test used.
expected	the mean of T under that null distribution.

References

Wu, Z. and Contador, G. (2026). Fisher's exact test controlling for confounders. Draft manuscript.

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Candes, E., Fan, Y., Janson, L. and Lv, J. (2018). Panning for gold: 'model-X' knockoffs for high dimensional controlled variable selection. *Journal of the Royal Statistical Society Series B*, 80(3), 551-577. doi:10.1111/rssb.12265

See Also

cfisher.null, fisher.test

Examples

```
set.seed(1)
n <- 300
z <- rnorm(n)                                # a confounder
x <- rbinom(n, 1, plogis(-1 + 1.2 * z))      # exposure depends on z
y <- rbinom(n, 1, plogis(-0.5 + z))          # outcome depends on z, not on x
fisher.test(table(x, y))$p.value             # ignores z
cfisher.test(x, y, z)                        # controls for z

# fit the null model once and test many exposures
fit <- glm(y ~ z, family = binomial)
cfisher.test(x, y, fit = fit, alternative = "greater", conf.int = FALSE)$p.value
```

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